

Beyond grades: what makes teachers retain students

Ignacio Garijo Campos

Antonio Alfonso

Pablo Brañas

Diego Jorrat

Ángel Sánchez

Universidad Loyola Andalucía

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Introduction: Grade Repetition

- Grade repetition can be seen as a **double punishment**: students face academic failure and are separated from their peers.
- This separation may negatively affect **self-esteem** and perceptions of ability.
- In psychology, this relates to **learned helplessness**: students stop trying because they believe success is impossible regardless of effort.
- Overall, the literature finds **little or no positive effects of grade repetition on test scores, skills or dropout rates**

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Our approach

- Our novel approach: we focus on teachers and their 'harshness' level
- We carry out a survey experiment in Spain, where the grade repetition rate more than doubles that of the OECD (9% and 22%)
- Double objective:
 - ① Characterizing 'harsh' teachers
 - ② Studying how teachers behave under different policy circumstances

Our approach: What is harshness?

Socioeconomic questionnaire → Experimental section → Behavioral questionnaire

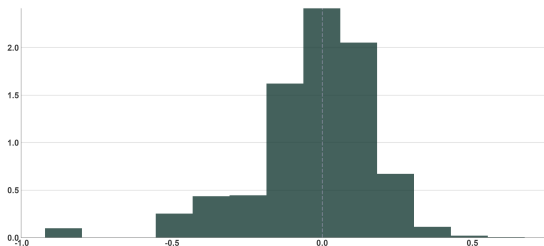
	Option 1	Option 2
Gender	Male	Female
Disadvantaged background	Yes	No
Failed 3 or more subjects	Yes	No
Lacks mathematical and/or linguistic abilities	Yes	No
Has committed a serious or very serious infraction or has been suspended	Yes	No
Frequently misses class without justification	Yes	No

Our approach: Measures of harshness

We estimate 4 different measures, but due to their similarity, only show results for one:

- 1 Number of repetitions
- 2 **Average deviation with respect to peers ($\simeq 320$)**
- 3 Average deviation with respect to predicted probability of repetition
- 4 Average deviation with respect to predicted probability of repetition in a more complete model

Figure: Average deviation with respect to peers



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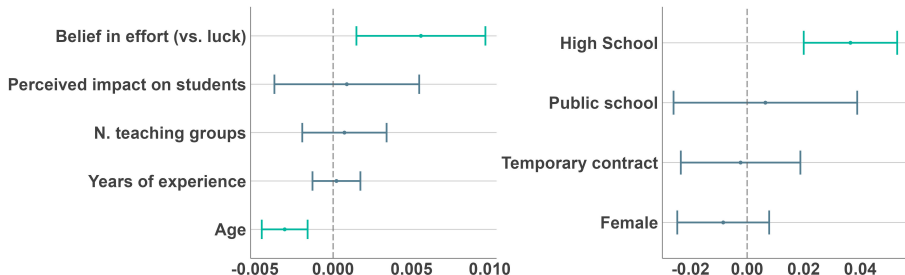
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Characterizing harsh teachers: the model

Figure: Coefficients of linear regression on harshness



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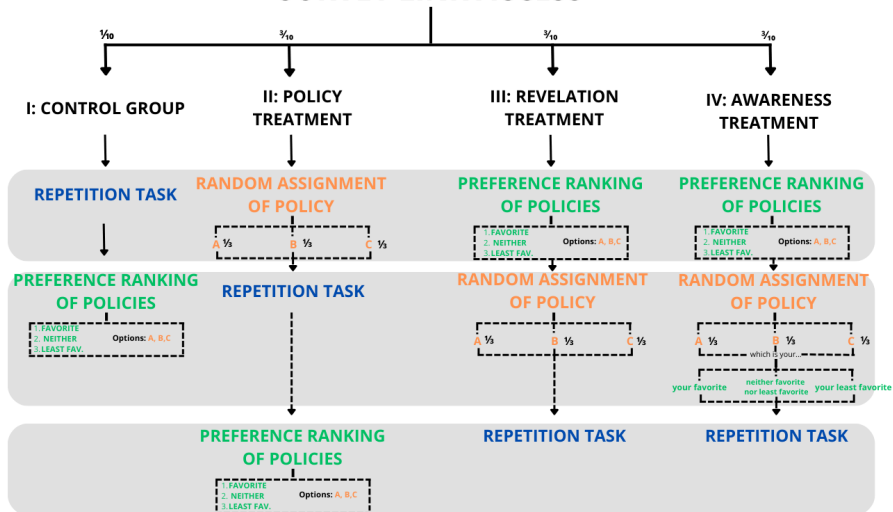
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The experiment: design

SURVEY LINK ACCESS



The experiment: does teacher behavior change when assigned to a policy?

	(1)	(2)
(Intercept)	0.014 (0.012)	0.023* (0.013)
assignedPolicy 1	-0.023 (0.017)	-0.025 (0.016)
assignedPolicy 2	0.017 (0.016)	0.019 (0.016)
assignedPolicy 3	-0.007 (0.016)	-0.001 (0.016)
favoritePolicy 2		0.046** (0.020)
favoritePolicy 3		-0.044*** (0.013)
Num.Obs.	1011	1011
R2	0.006	0.028
F	2.034	5.833

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The experiment: does teacher behavior change when assigned to a policy they like/dislike?

	(1)	H13
(Intercept)	0.014 (0.012)	0.021* (0.013)
assignmentfavorite	-0.015 (0.016)	-0.011 (0.016)
assignmentleast-favorite	0.005 (0.016)	0.007 (0.016)
favoritePolicy 2		0.064*** (0.022)
favoritePolicy 3		-0.044*** (0.014)
Num.Obs.	769	769
R2	0.002	0.032
F	0.875	6.293

* p < 0.1, ** p < 0.05, *** p < 0.01

The experiment: does teacher behavior change when assigned to a policy they already said they like/dislike?

	Favorite	Least favorite
(Intercept)	−0.004 (0.013)	0.040*** (0.014)
DRevelation treatment	0.013 (0.016)	−0.022 (0.017)
favoritePolicy 2	0.095*** (0.029)	0.043 (0.029)
favoritePolicy 3	−0.013 (0.017)	−0.074*** (0.018)
Num.Obs.	530	483
R2	0.026	0.054
F	4.718	9.095

* p > 0.1, ** p > 0.05, *** p > 0.01

The experiment: does teacher behavior change when assigned to a policy they are being reminded they do not like?

	Favorite	Least favorite
(Intercept)	0.013 (0.014)	0.003 (0.015)
DAwareness treatment	-0.022 (0.016)	0.001 (0.018)
favoritePolicy 2	0.089*** (0.030)	0.060* (0.032)
favoritePolicy 3	-0.023 (0.017)	-0.041** (0.018)
Num.Obs.	551	497
R2	0.027	0.023
F	5.076	3.800

* p > 0.1, ** p > 0.05, *** p > 0.01

So... Did anything change at all?

Teachers were more (less) likely to say their (least) favorite policy was Policy 3 when assigned to it

	Least favorite			Favorite		
	Policy 1	Policy 2	Policy 3	Policy 1	Policy 2	Policy 3
(Intercept)	-3.065*** (0.213)	1.086*** (0.102)	-1.228*** (0.107)	0.179** (0.088)	-2.213*** (0.149)	-0.779*** (0.096)
assigned	0.469 (0.330)	-0.099 (0.174)	-0.372* (0.197)	0.109 (0.156)	-0.002 (0.258)	0.424*** (0.158)
Num.Obs.	761	761	761	761	761	761

* p $\geq$ 0.1, ** p $\geq$ 0.05, *** p $\geq$ 0.01

- Grade repetition is not only a matter of students! Who is the teacher matters
- Teacher's gender, experience, contract situation or perceived positive impact on students does not matter; **belief in effort, age and grade level does**
- Teachers do not change their grade repetition patterns when assigned to a policy
- Teachers' alignment with the policy does not change their grade repetition patterns either
- Mere exposure to a policy may have an impact on teacher's preferences